

James Mba Azam

Curriculum Vitae

London School of Hygiene and
Tropical Medicine (LSHTM),
Keppel Street,
London WC1E 7HT,
United Kingdom.
☎ (+44) 7931 424395
✉ james.azam@lshtm.ac.uk

Education

- Feb, 2018 – April, 2022 **PhD Applied Mathematics**, DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA), Stellenbosch University, South Africa
Thesis *Modelling outbreak response intervention strategies for decision-making*
- Mar, 2016 – Mar, 2018 **MSc Mathematics**, DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA), Stellenbosch University, South Africa
Thesis *Assessment of a measles outbreak response vaccination campaign, and two measles parameter estimation methods*
- Aug, 2014 – Jun, 2015 **Structured MSc Mathematical Sciences**, African Institute for Mathematical Sciences (AIMS), Senegal
Thesis *Hedging Conditional Value at Risk*
- Aug, 2009 – Jun, 2013 **BSc Actuarial Science**, Kwame Nkrumah University of Science and Technology, Ghana
Thesis *Predicting optimal periods for the sale of cocoa in Ghana for the year 2013, using futures contracts*

Recent Publications

- Azam, J. M.**, Pang, X., Are, E. B., Pulliam, J. R., & Ferrari, M. J. (2023). Modelling outbreak response impact in human vaccine-preventable diseases: A systematic review of differences in practices between collaboration types before COVID19. *Epidemics*, 100720.
- Kent, B. A., Holman, C., Amoako, E., Antonietti, A., **Azam, J. M.**, Ballhausen, H., ... & Weissgerber, T. L. (2022). Recommendations for empowering early career researchers to improve research culture and practice. *PLoS biology*, 20(7), e3001680.
- Azam, J.M.**, Saitta, B., Bonner, K., Ferrari, M.J. and Pulliam, J.R. (2021). Modelling the relative benefits of using the measles vaccine outside cold chain for outbreak response. *Vaccine*. DOI: 10.1016/j.vaccine.2021.08.053

Professional Positions

- Nov, 2022 – Current **Research Software Engineer**, Epiverse Initiative, London School of Hygiene and Tropical Medicine, London, United Kingdom
- Feb, 2022 – Nov, 2022 **Junior Researcher**, DSI-NRF Center of Excellence in Epidemiological Modelling and Analysis, Stellenbosch, South Africa
- Feb, 2021 – Sep, 2021 **Data Science Consultant (Part-time)**, *Winning Insights and Models for Marvellous Yields (Wimpy)*, Cape Town, South Africa

Fellowships and Awards

Oct, 2018 **Big Data Policy Fellowship, £4000**, Development in Africa with Radio Astronomy (DARA), University of Manchester, United Kingdom
A one month fellowship in Science and Policy Communication for Big Data projects.

Technical Competencies

Programming Advanced R (package development), Bash, GNU make, Github & Git

Open Source Contributions

Developer & maintainer `epichains` & `bpmodels` - R packages for simulating and analysing transmission chains simulated from branching process models.

Co-author `epinowcast` - An R package for flexible and efficient hierarchical nowcasting of right-truncated epidemiological time-series using a semi-mechanistic Bayesian model with support for a range of reporting and generative processes.

Co-author `EpiNow2` - An R package for estimating real-time case counts and time-varying epidemiological parameters.

Co-maintainer Data Carpentry lesson: "R for data analysis and visualization of Ecological Data"

Conferences

Oral Presentations

Aug, 2023 Superseding R packages – lessons learned from transitioning to the `epichains` R package from `bpmodels`. *RSLondonSouthEast, Imperial College, London, United Kingdom*

Nov, 2021 Modelling the relative benefits of using the measles vaccine outside cold chain for outbreak response. *Epidemics8, Elsevier (Virtual presentation)*

Workshops Attended

Aug, 2023 **RSLondonSouthEast 2023**, *Imperial College London, London, United Kingdom*

Mar, 2023 **Forecasting natural and social systems**, *The Royal Society, London, United Kingdom*

Dec, 2022 **100 days and 100 lines of code**, *Epiverse Initiative, London School of Hygiene and Tropical Medicine, London, United Kingdom*

Jan, 2020 **Software Engineering for Applied Mathematical Sciences**, *SACEMA, Stellenbosch*

Reviewer

○ Journey of Open Science Software ○ PLOS Global Health

Diversity, Equity, and Inclusion

Dec, 2022 – Oct 2023 **Active Member**, *Equity, Diversity, Inclusion, and Accessibility Working Group*, Society of Research Software Engineering, United Kingdom

Jun, 2023 - Sept, 2023 **Career Panel Member**, *HBR-UK Black Internship Programme*, London School of Hygiene and Tropical Medicine, United Kingdom